Day 2 - 4

Semantic segmentation

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Schedule

Each hour is scheduled as **50 min lecture + 10 min break**

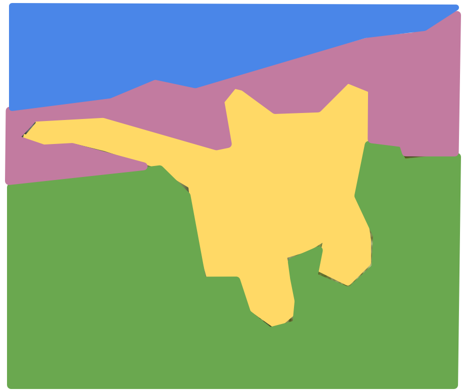
- **Day 1**: 4 hr lecture + 4 hr practice
  - **AM**: CNN, Transformers, Object detection
  - **PM**: DETR, MI-DETR
- **Day 2**: 4 hr lecture + 4 hr practice
  - **AM**: Self-supervised learning, Weakly supervised learning, Multimodal models, Semantic segmentation
  - **PM**: DINO, SAM variants

Computer Vision Tasks

Classification

CAT

Semantic Segmentation

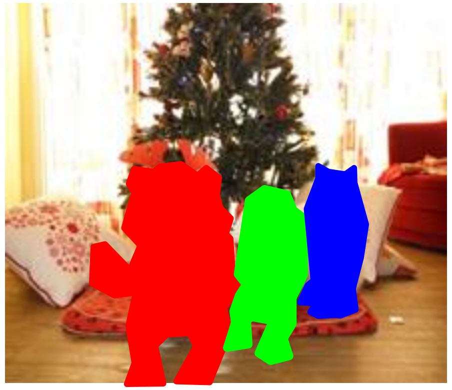


GRASS, CAT, TREE, SKY

Object Detection

DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

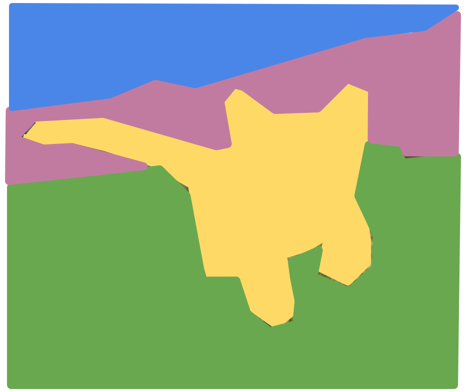
No spatial extent

No objects, just pixels

Multiple Objects

This image is CC0 public domain

Tasks: Semantic Segmentation

Classification	**Semantic Segmentation**	Object Detection	Instance Segmentation
			
CAT	GRASS, CAT, TREE, SKY	DOG, DOG, CAT	DOG, DOG, CAT
No spatial extent	No objects, just pixels	Multiple Objects	

Semantic Segmentation

Label each pixel in the image with a category label

Don't differentiate instances, only care about pixels

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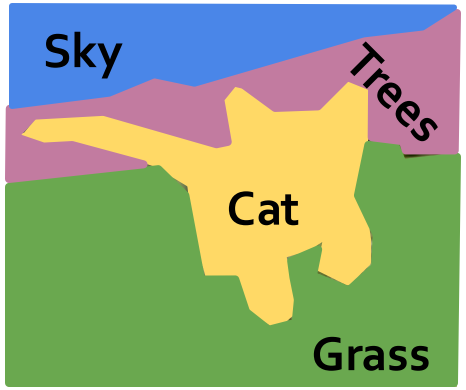
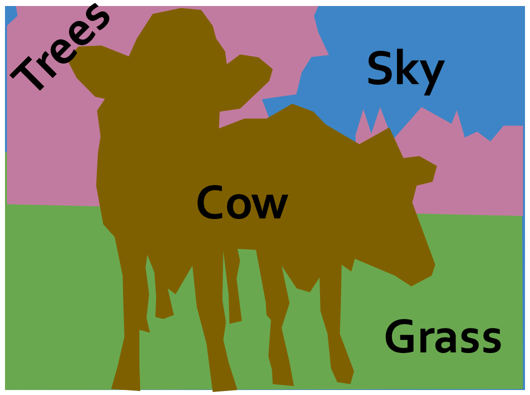
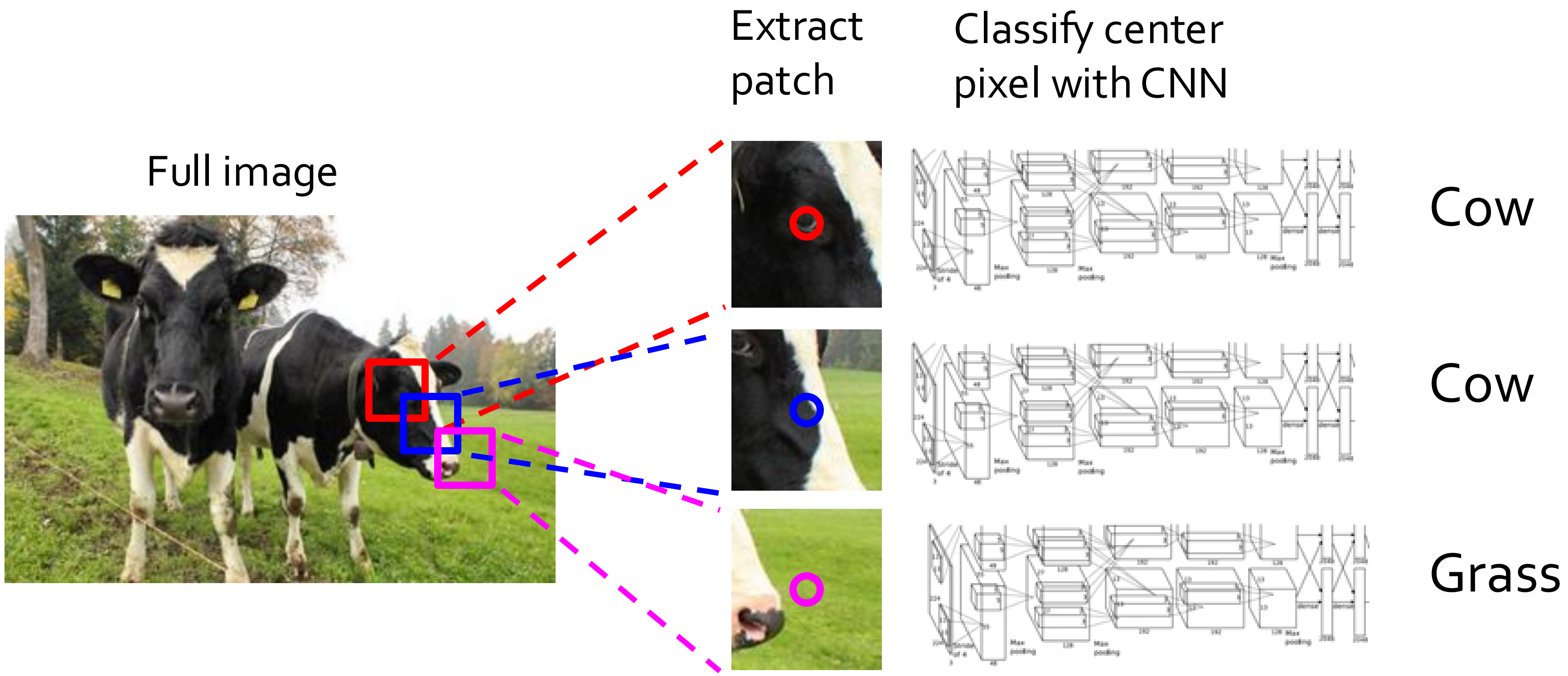




Segmentation: Sliding Window



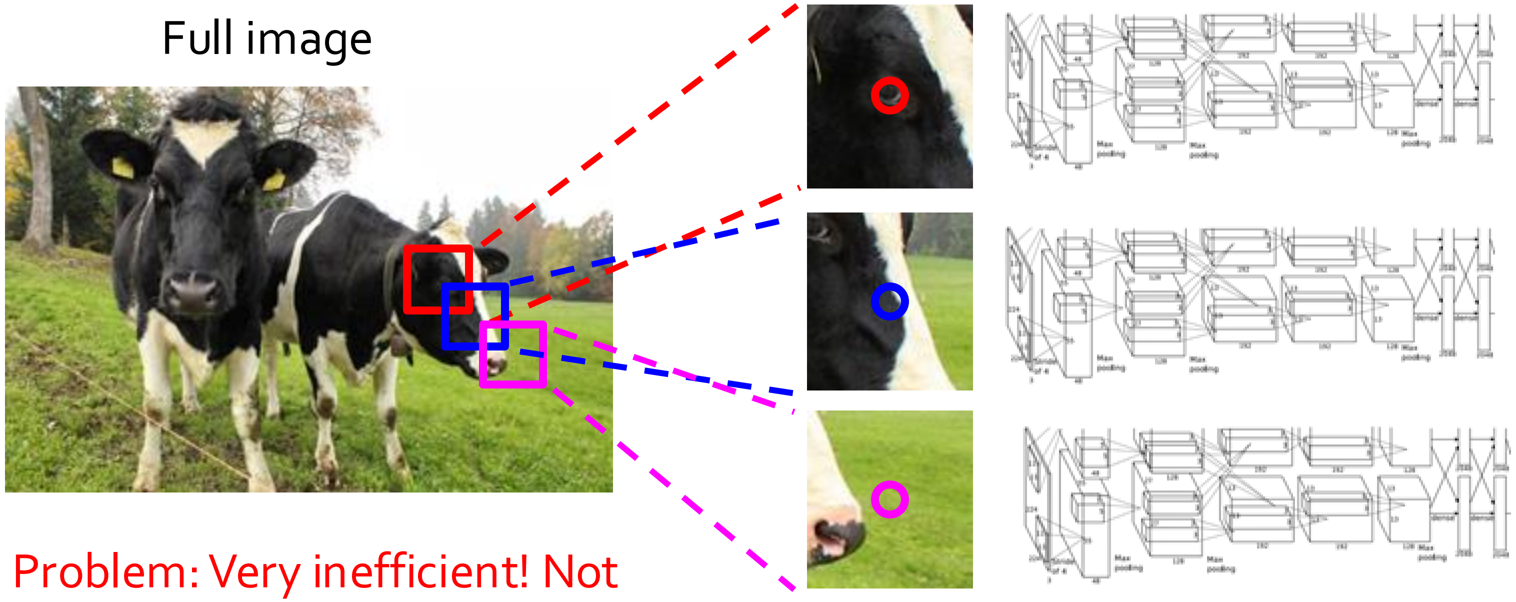
Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Segmentation: Sliding Window

Full image

Extract patch

Classify center pixel with CNN



Cow

Cow

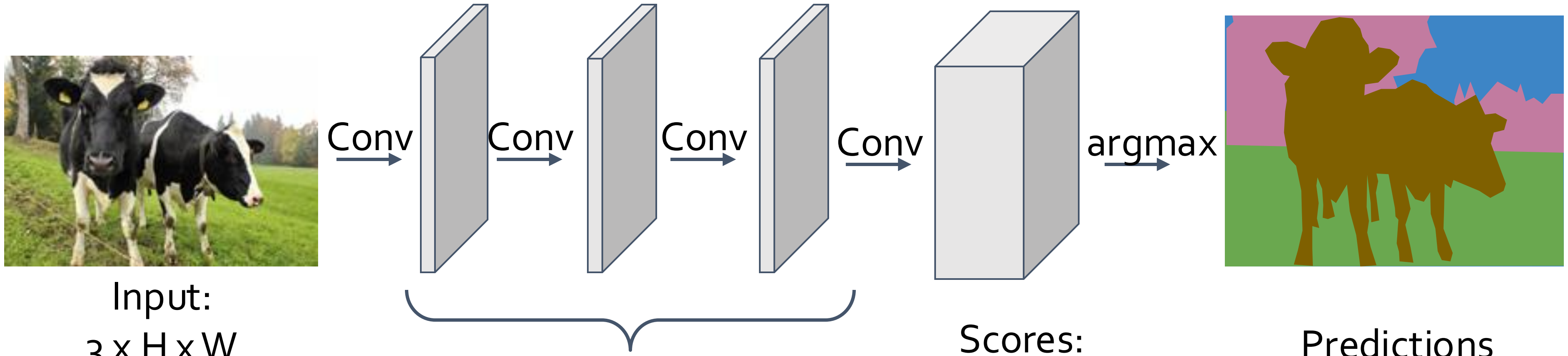
Grass

Problem: Very inefficient! Not reusing shared features between overlapping patches

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Input: 3 x H x W

Convolutions: D x H x W

Scores: C x H x W

Predictions : H x W

Loss function: Per-Pixel cross-entropy

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Fully Convolutional Network

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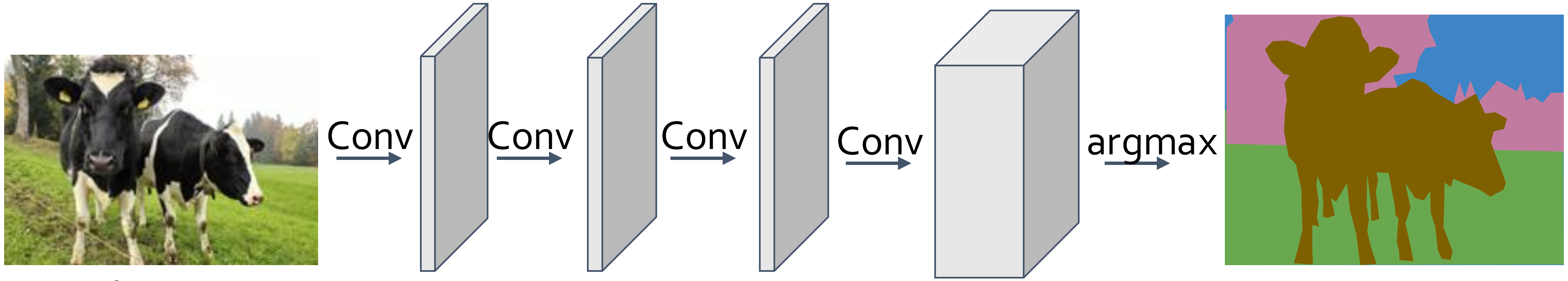


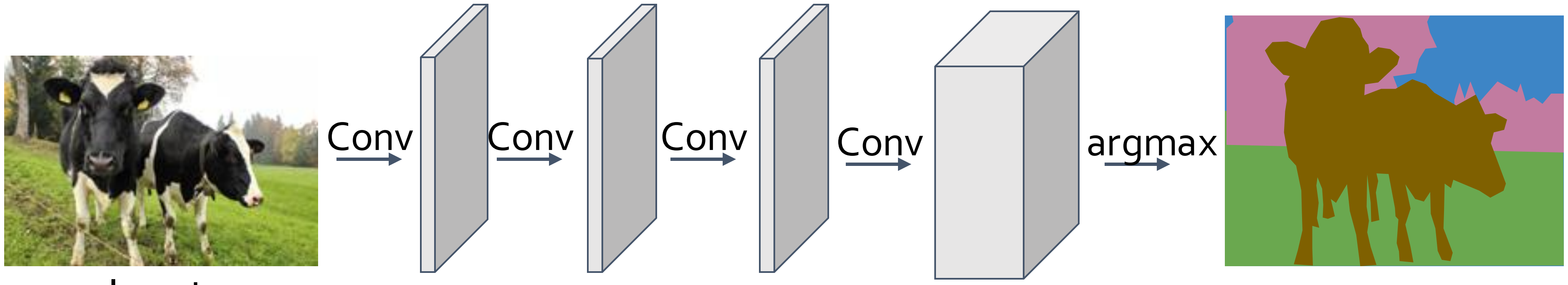
Input:
3 x H x W

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3x3 conv layers, receptive field is 1+2L

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Input:
3 x H x W

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3x3 conv layers, receptive field is 1+2L

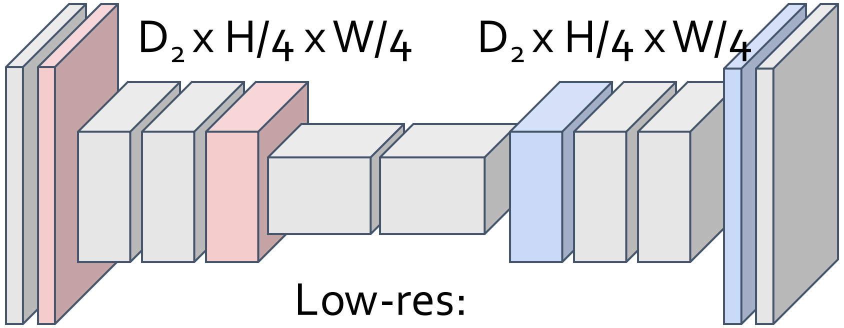
Problem #2: Convolution on high res images is expensive!

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Fully Convolutional Network

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
$3 \times H \times W$



Med-res:
$D_2 \times H/4 \times W/4$

Med-res:
$D_2 \times H/4 \times W/4$

High-res:
$D_1 \times H/2 \times W/2$

Low-res:
$D_3 \times H/4 \times W/4$

High-res:
$D_1 \times H/2 \times W/2$

Predictions:
$H \times W$

Downsampling:
Pooling, strided convolution

Upsampling:
???

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

In-Network Upsampling: “Unpooling”

Bed of Nails

Input
C x 2 x 2

1	2
3	4

Output
C x 4 x 4

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Nearest Neighbor

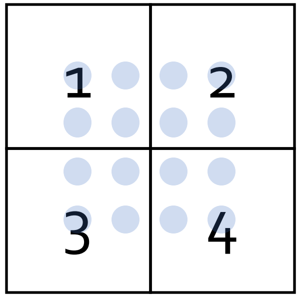
Input
C x 2 x 2

1	2
3	4

Output
C x 4 x 4

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Upsampling: Bilinear Interpolation



Input: C x 2 x 2

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

Output: C x 4 x 4

$$f_{x,y} = \sum_{i,j} f_{i,j} \max(0, 1 - |x - i|) \max(0, 1 - |y - j|) \quad i \in \{\lfloor x \rfloor - 1, \dots, \lceil x \rceil + 1\}$$

$$j \in \{\lfloor y \rfloor - 1, \dots, \lceil y \rceil + 1\}$$

Use two closest neighbors in x and y to construct linear approximations

Fully Convolutional Network

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
$3 \times H \times W$

Med-res:
$D_2 \times H/4 \times W/4$

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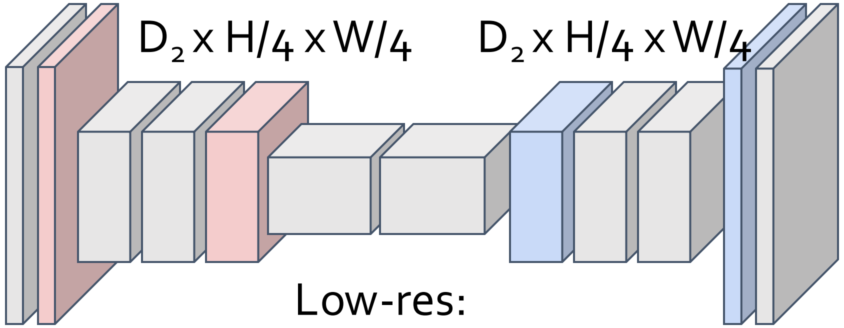
$D_2 \times H/4 \times W/4$



High-res:
$D_1 \times H/2 \times W/2$

Low-res:
$D_3 \times H/4 \times W/4$

High-res:
$D_1 \times H/2 \times W/2$

Predictions:
$H \times W$

Downsampling:
Pooling, strided convolution

Upsampling:
???

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Tasks: Instance Segmentation

Classification

CAT

No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SKY

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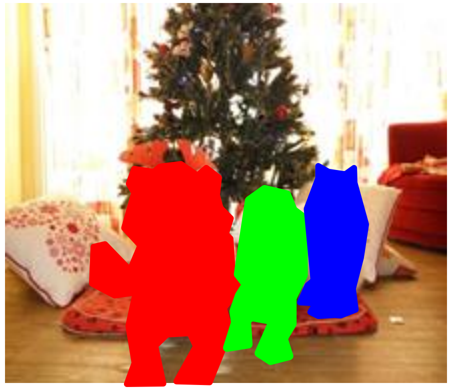
Object Detection

DOG, DOG, CAT

Instance Segmentation

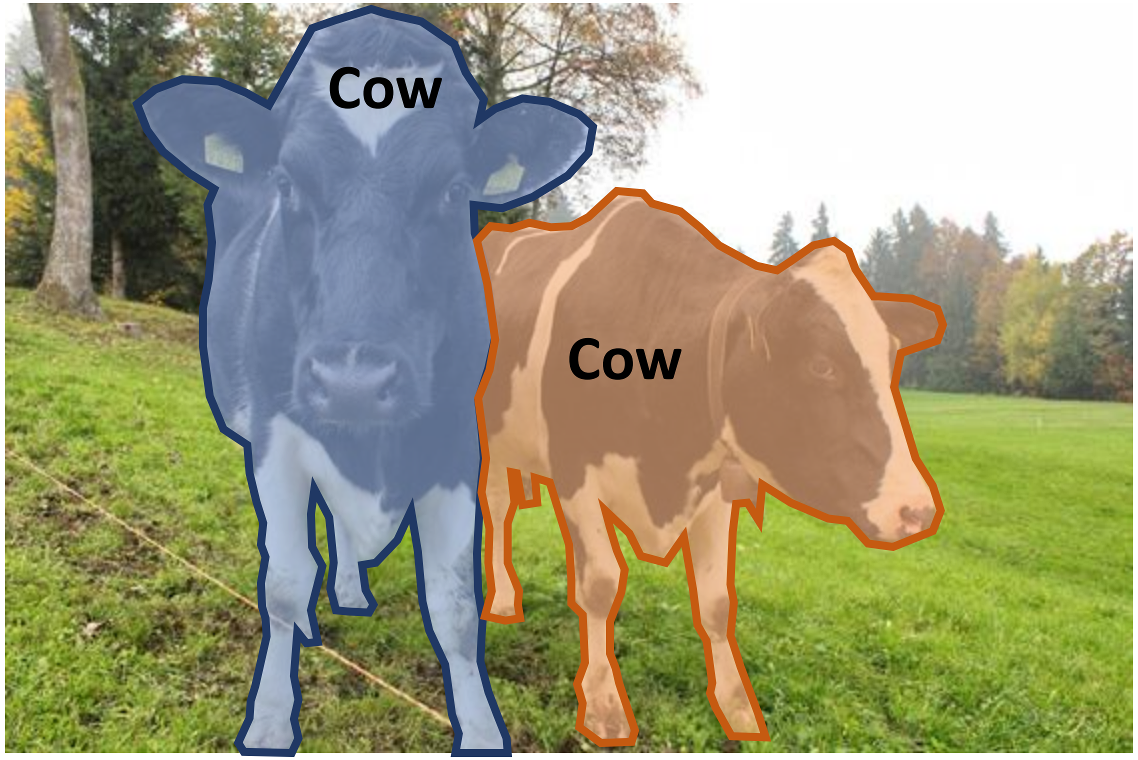


DOG, DOG, CAT

Multiple Objects

Instance Segmentation

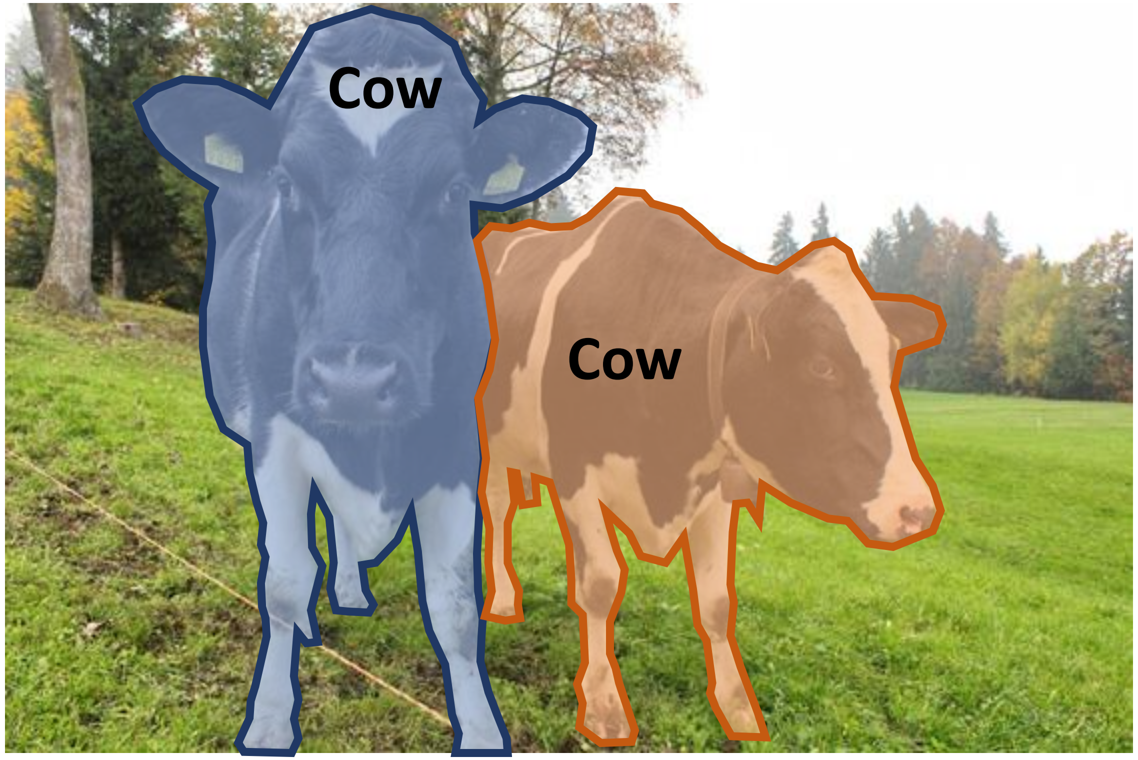
- **Instance Segmentation**: Detect all objects in the image, and identify the pixels that belong to each object



This image is CC0 public domain

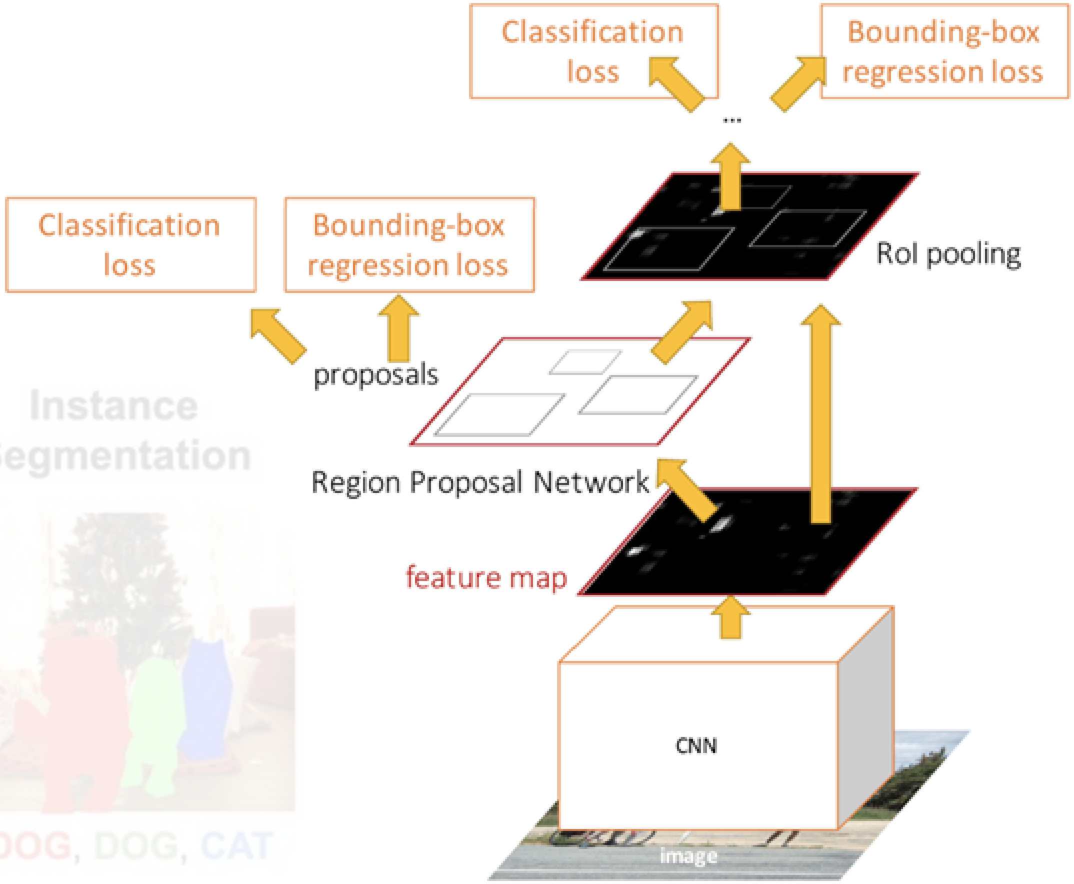
Instance Segmentation

- **Instance Segmentation**: Detect all objects in the image, and identify the pixels that belong to each object
- **Approach:** Perform object detection, then predict a segmentation mask for each object!



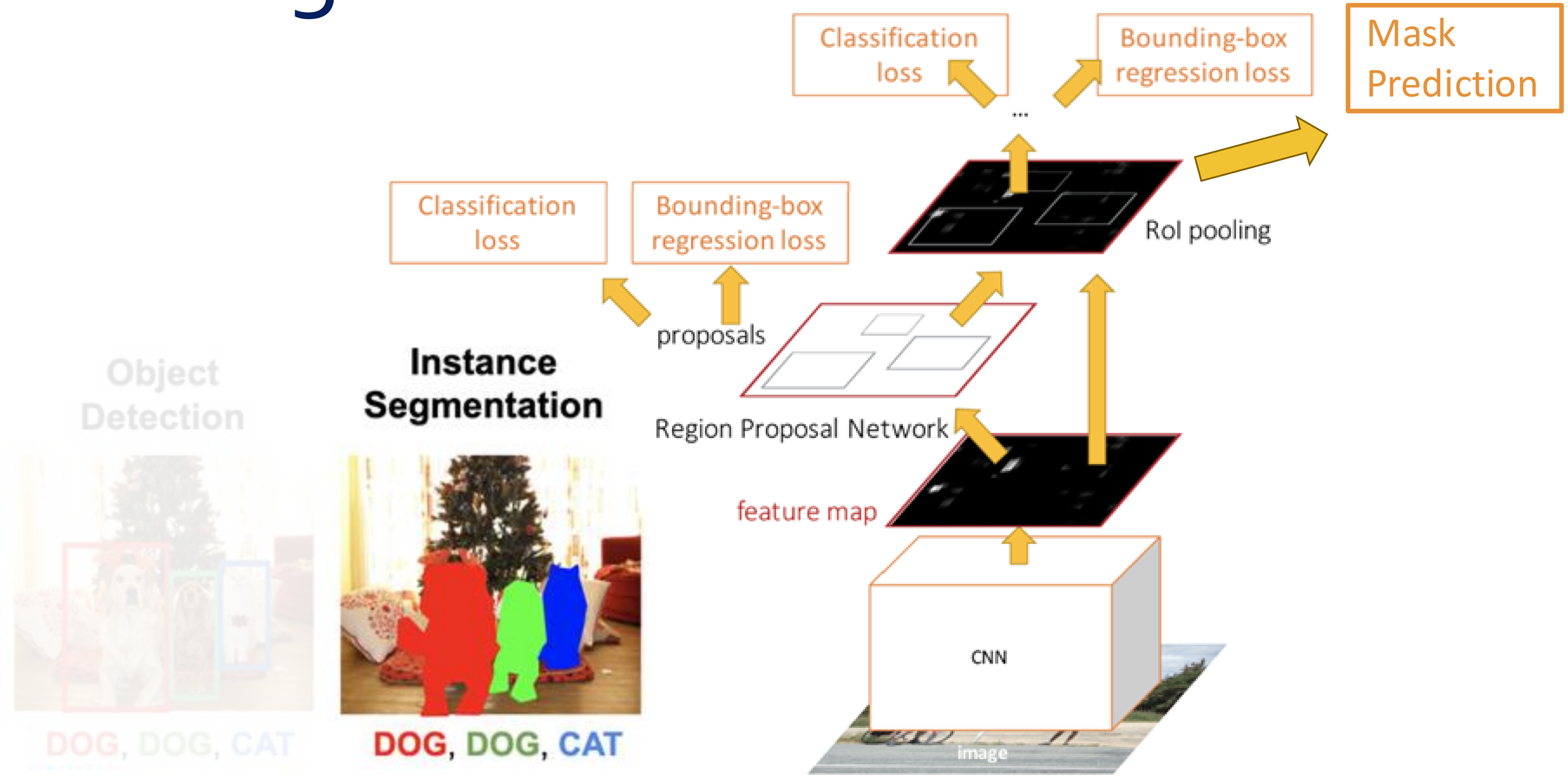
This image is CC0 public domain

Object Detection: Faster R-CNN



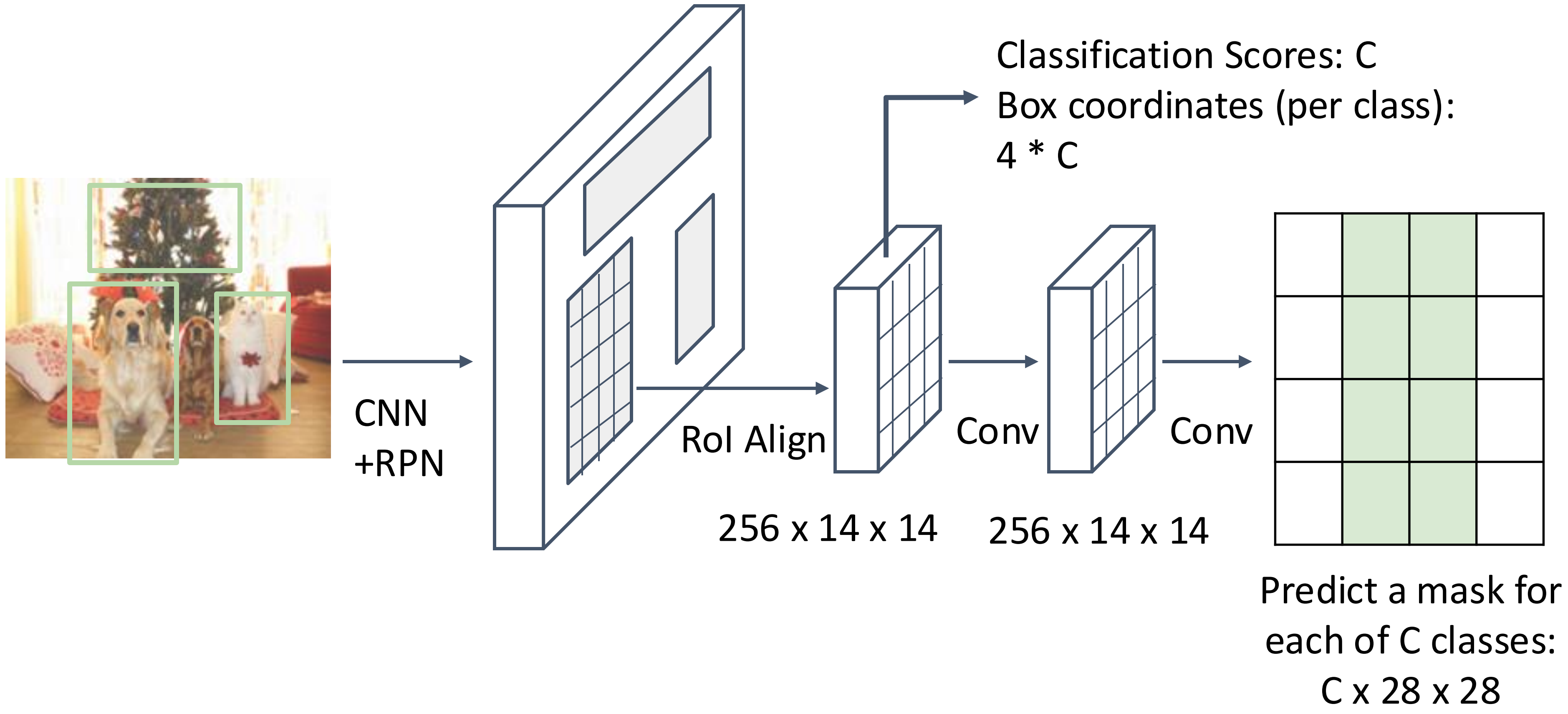
Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NeurIPS 2015

Instance Segmentation: **Mask R-CNN**



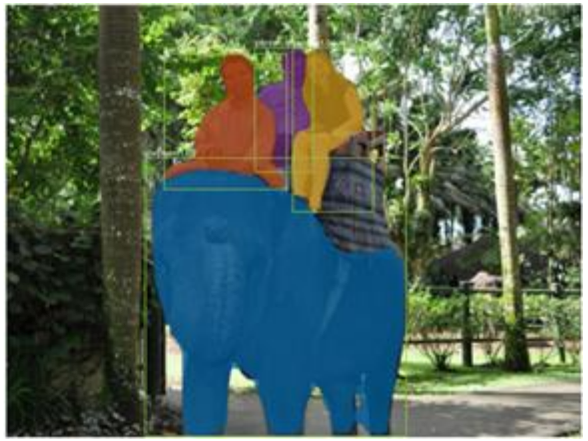
He et al, "Mask R-CNN", ICCV 2017

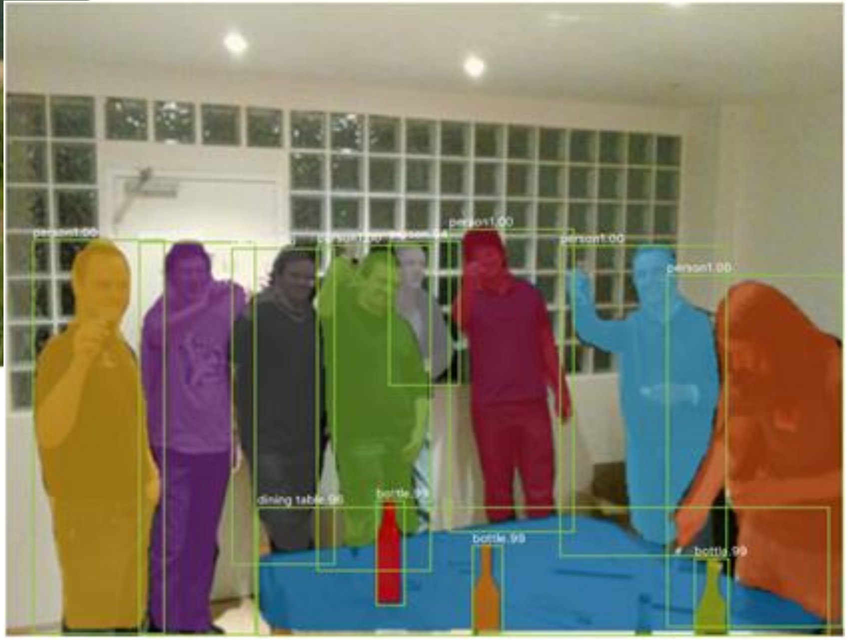
Mask R-CNN



He et al, "Mask R-CNN", ICCV 2017

Mask R-CNN: Very Good Results!





He et al, “Mask R-CNN”, ICCV 2017

SAM: Segment Anything

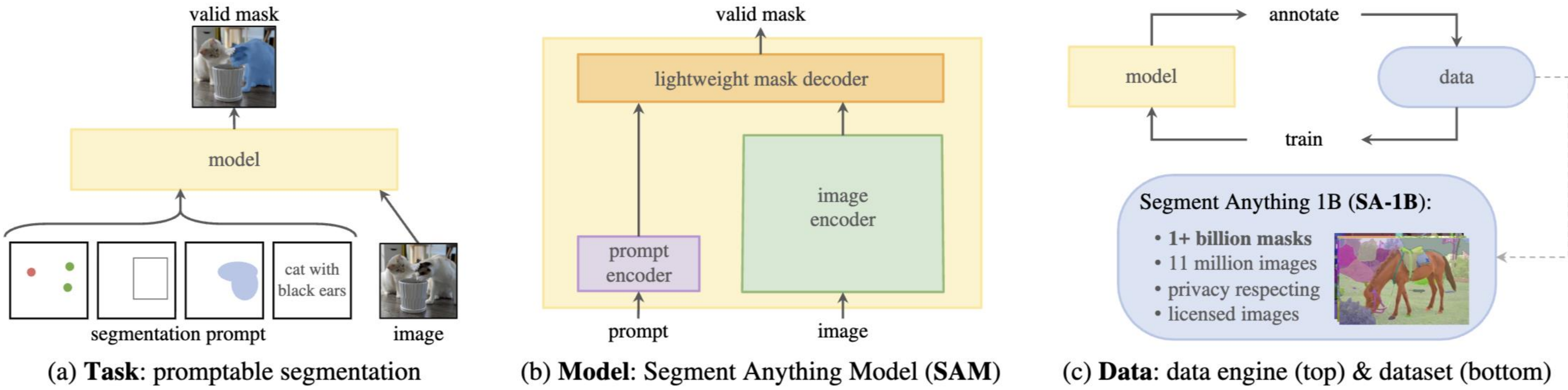


Figure 1: We aim to build a foundation model for segmentation by introducing three interconnected components: a promptable segmentation *task*, a segmentation *model* (SAM) that powers data annotation and enables zero-shot transfer to a range of tasks via prompt engineering, and a *data* engine for collecting SA-1B, our dataset of over 1 billion masks.

Kirillov *et al.*, **Segment Anything**, *ICCV 2023*

SAM: Segment Anything

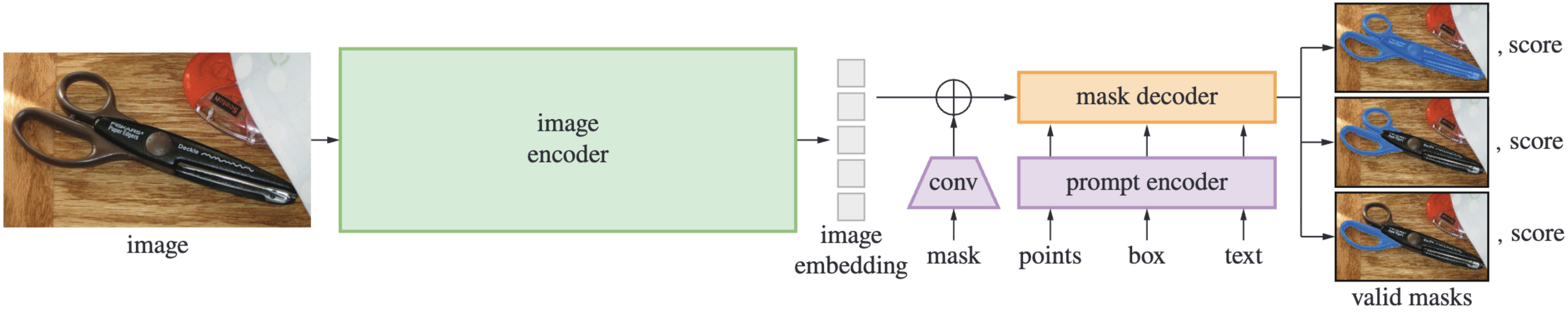


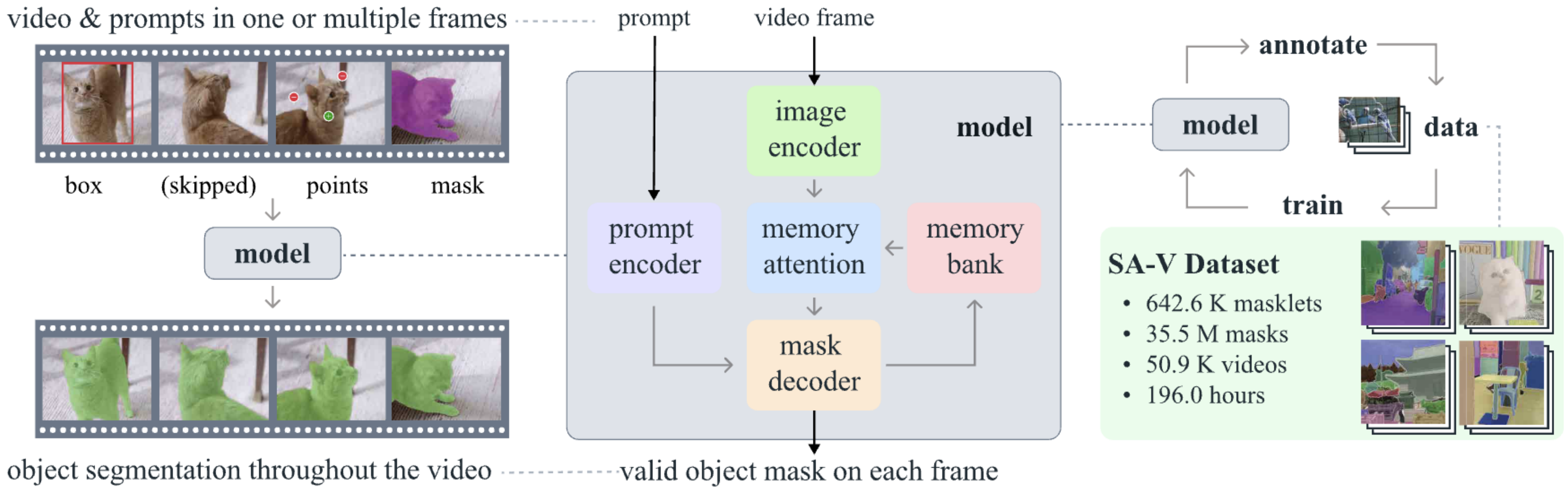
Figure 4: Segment Anything Model (SAM) overview. A heavyweight image encoder outputs an image embedding that can then be efficiently queried by a variety of input prompts to produce object masks at amortized real-time speed. For ambiguous prompts corresponding to more than one object, SAM can output multiple valid masks and associated confidence scores.

Kirillov *et al.*, **Segment Anything**, *ICCV 2023*

SAM: Segment Anything

Kirillov *et al.*, **Segment Anything**, *ICCV 2023*

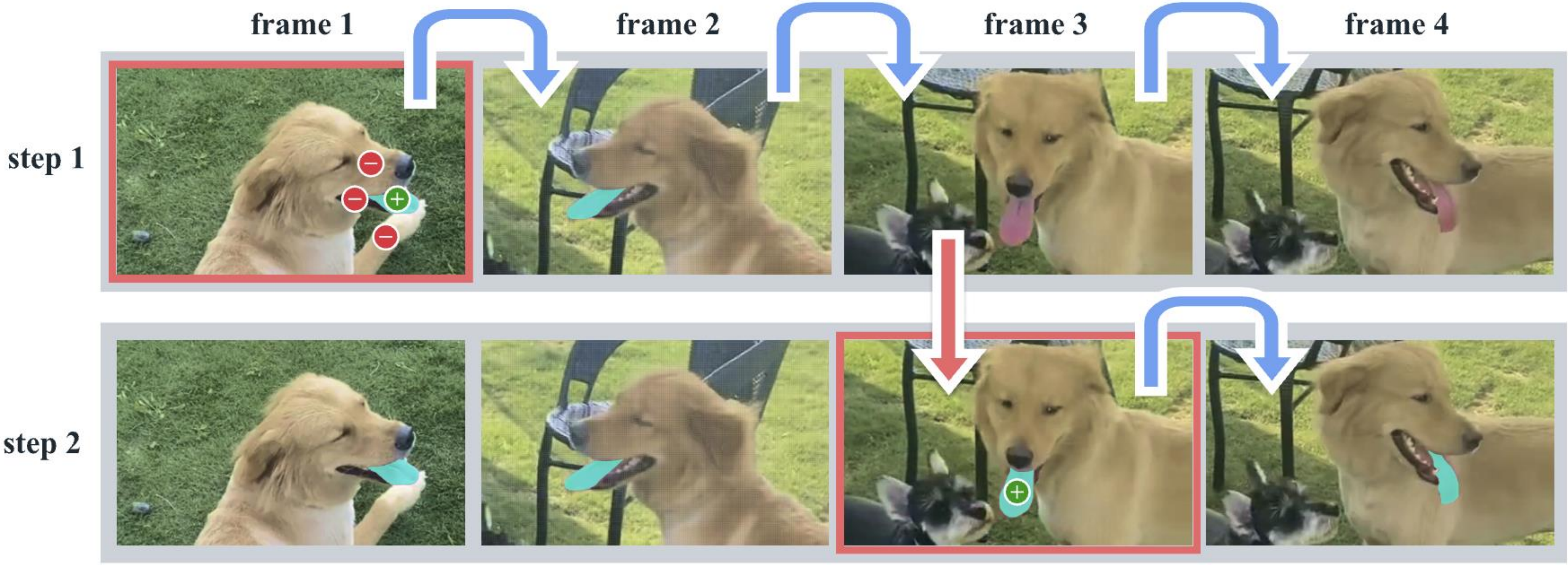
SAM 2: Segment Anything in Images and Videos



(a) Task: promptable visual segmentation **(b) Model:** Segment Anything Model 2 **(c) Data:** data engine and dataset

Ravi *et al.*, **SAM 2: Segment Anything in Images and Videos**, *ICLR 2025 (Oral) – Outstanding paper honorable mention award*

SAM 2: Segment Anything in Images and Videos



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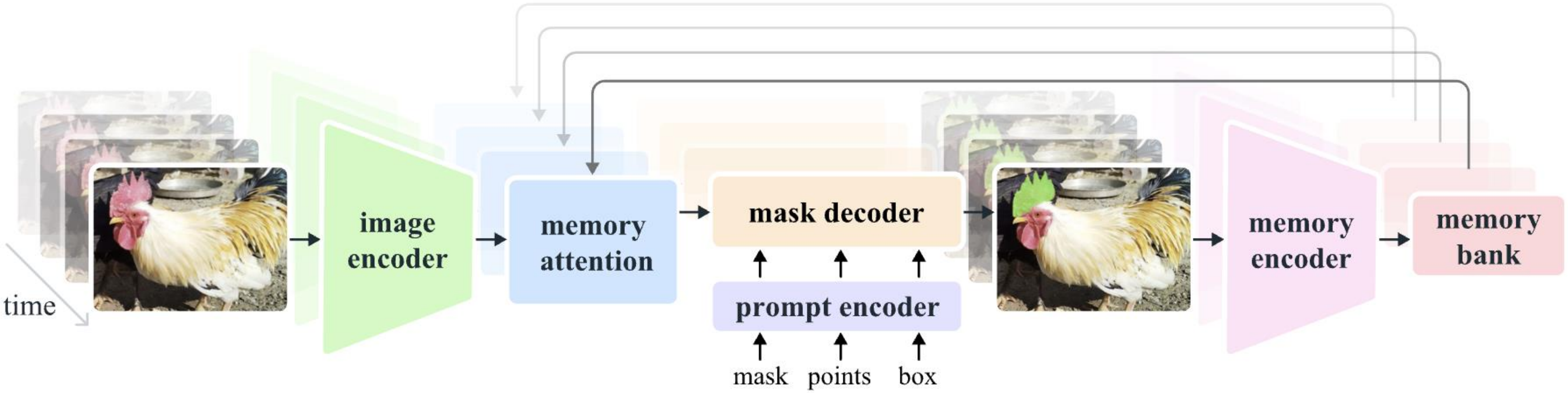

Ravi *et al.*, **SAM 2: Segment Anything in Images and Videos**, *ICLR 2025 (Oral) – Outstanding paper honorable mention award*

SAM 2: Segment Anything in Images and Videos

Figure 4: Example videos from the SA-V dataset with masklets overlaid (manual and automatic). Each masklet has a unique color, and each row represents frames from one video, with 1 second between them. More examples in Fig. 11.

Ravi *et al.*, **SAM 2: Segment Anything in Images and Videos**, *ICLR 2025 (Oral) – Outstanding paper honorable mention award*

Summary

- Few-shot learning
- Semi-supervised learning
- Active learning